

4	a(i)	X, Z, Y. (Explanation: The brightness of the bulbs depends on the power dissipated in them. Comparing bulbs X and Z, since they are in parallel, the voltage drop across them is the same, and using $P = V^2/R$, we conclude that bulb Z is brighter than bulb X. Next, use $P = I^2R$ to compare between bulbs Y and Z, and bulb Y is brighter as the current through Y is larger.)	A1
	(ii)	$R_{eff} = \left(\frac{1}{4.0} + \frac{1}{2.0} \right)^{-1} + 2.0$ $= 3.33$ $= 3.3 \Omega$	C1 A1
	(iii)	$V_{AB} = \frac{R_{eff}}{R_{eff} + r} \times E$ $= \frac{3.33}{3.33 + 0.5} \times 3.0$ $= 2.608$ $= 2.6 \text{ V}$	C1 A1
	b(i)	Using the potential divider principle on the upper circuit, $V_{CD} = 6.0 \frac{2.0}{1.0 + 2.0}$ $= 4.0 \text{ V}$	M1 A0
	(ii)	At null deflection, $V_{CE} = V_{AB} = 2.608 \text{ V}$ $\frac{L_{CE}}{L_{CD}} = \frac{V_{CE}}{V_{CD}}$ $\frac{L_{AC}}{100.0} = \frac{2.608}{4.0}$ $L_{CE} = 65.2 = 65 \text{ cm}$	C1 A1
	(iii)	With a lower resistance bulb used for X, the <u>effective resistance</u> of the three bulbs (a(ii)) will <u>decrease</u>, hence the <u>terminal potential difference (a(iii))</u> across the cell in the lower circuit <u>decreases</u>. Therefore, the balance length AC must be <u>shorter</u> for null deflection to occur.	M1 A1
		Total:	10